



UNIVERSITÀ DEGLI STUDI DI GENOVA

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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

RESEARCH TRACK 2

Third Assignment

Statistical Analysis

Author:

Amirmahdi Matin

Student ID:

s5884715

Professors:

Carmine Recchiuto

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1 Introduction

A comprehensive statistical analysis is carried out on the first assignment from Research Track 1 as part of the final work of Research Track 2. Simulating and directing a mobile robot in an area was the first task that must be completed for Research Track 1. It was expected that the robot would collect all of the golden boxes that were arranged close to each other in a single location inside the environment. Within the scope of this report, a comparison is made between the performance of my code and the performance of someone else's code that does the same task. Using the hypotheses that are discussed in the next section, the evaluation of performance is carried out. Both the amount of time it takes for the robot to collect all of the golden boxes that are adjacent to one another and the number of times that each algorithm is successful and unsuccessful in detecting and collecting the golden boxes when the number of boxes in the environment changes are the parameters that determine which algorithm performs better.

2 Hypothesis

The statistical method known as hypothesis testing is typically utilized in the field of scientific research in order to arrive at a conclusion regarding the parameters of a data set (such as the mean, variance, and so on).

On the basis of two competing hypotheses, the hypothesis for a particular situation might be described as follows: The null hypothesis (H_0) asserts that the two procedures that are being compared are of identical quality (in accordance with the evaluation methods that will be utilized subsequently). On the other hand, the alternative hypothesis, often known as H_a , is something that is defined as the assertion that one approach is superior to the other method. Both of these hypotheses are false if one of them is true, and vice versa if the other one becomes true. The rejection of the null hypothesis is one way that can be used to demonstrate that the alternative hypothesis is supported by evidence and that one method is superior to the other method. On the basis of this, the null and alternative hypotheses should be defined in the following manner as the initial stage in the process of assessing the two codes contained in this report:

2.1 Null Hypothesis

Both the results produced from my code and the results got from the code of my colleague have the same performance capabilities and are comparable to one another. The average amount of time required to complete this work is the same for both algorithms (i.e., $\mu_A = \mu_B$), which means that we are unable to tell which algorithm is more efficient than the other during the process of gathering the boxes.

2.2 Alternative hypothesis

When it comes to grouping the boxes next to each other under a variety of conditions, the algorithm written by my colleague is far faster than my own approach. Consequently, the average amount of time that my robot takes to collect all of the boxes is longer than the amount of time that my colleague's code takes (i.e., μ_A is greater than μ_B).

2.3 Tests

The first thing that has to be done in order to test the null hypothesis is to collect some sample test data. The amount of data collected throughout the exam ought to be sufficient in order to produce results that are reliable, accurate, and appropriate. The Z-test, the T-test, the two sampled T-test, the paired T-test, and other similar tests are some of the various approaches that can be utilized to examine and compare the performance of two algorithms. In order for us to be able to proceed with the analysis using a certain approach, there are a few fundamental preconditions that must be met. Typically, the Z-test is utilized in situations when the population, as well as its mean and standard deviation, are accessible. When using the T-test, it is necessary to have knowledge about at least one of the characteristics of the population, such as the mean or the standard deviation. For the purpose of this research, the population is not available; rather, we are merely contrasting two distinct algorithms that are capable of completing a task. The paired T-test and the two sampled T-test are both potential usefulnesses for this analysis.

3 Experiment

3.1 Theory

It is necessary to establish identical environments for both algorithms in order to fulfill the need of obtaining test data for the analysis. Each algorithm is put through its paces in a variety of scenarios, with 4, 6, and 8 golden boxes, in order to see how quickly it is able to collect all of the boxes and how accurately it can identify the boxes and collect them. The algorithms are run fifteen times in each environment setup, which results in a total of forty-five simulations for each algorithm. We did not employ random creation of the boxes since we wanted to ensure that each technique was implemented in the exact same setting and that the comparison was accurate. As a consequence of this, the placements of the boxes remain the same in an environment that contains the same quantity of boxes. It has been agreed that a paired T-test would be performed on the data that was extracted. When we simulated both algorithms 45 times with the same amount of randomly generated golden boxes in the environment, we could have also used a two-sample T-test. This would have allowed us to obtain the same results.

The following is the recommended implementation of this method:

- The difference between the two observations for each simulation pair of two algorithms should be calculated using the formula $d_i = y_i - x_i$. It is important to differentiate between positive and negative differences while doing this calculation.
- The mean difference ($\bar{d}$) should be calculated.
- The standard error of the mean difference can be calculated by calculating the standard deviation of the differences, which is denoted by s_d . The formula for calculating the standard error of the mean difference is $SE(\bar{d}) = \frac{s_d}{\sqrt{n}}$.
- Determine the t-statistic, denoted as T, using the formula $T = \frac{\bar{d}}{SE(\bar{d})}$. When the null hypothesis is assumed to be true, this statistic conforms to a t distribution with $n - 1$ degrees of freedom.
- Utilize t-distribution tables to compare your calculated value for T with the t_{n-1} distribution. This will calculate the p-value for the paired t-test.

3.2 Results

The results of my code and the code simulation time and success of my colleague are provided in the following table under a variety of various conditions. Upon closer inspection, it is evident that both codes were totally successful in locating and collecting the boxes in each and every one of the cases that were studied. In some instances where there were only a few gold boxes, my code initially misplaced one of the boxes; but, later on in the simulation, the same box was picked up once more and placed near to other boxes. Due to the fact that the robot has ultimately gathered all of the boxes adjacent to each other, which is the sole requirement for determining whether or not the algorithm was successful, this particular instance cannot be called a failure. In spite of this, the simulation duration of the algorithm is finally impacted by the fact that the robot has picked up and moved a box over twice. As a consequence of this, the impact of this very minor error is projected in the computation of the algorithms' performance evaluation.

procedure The paired T-test value, as determined from the observations and calculations in the Table, is precisely 15.35. The data is subjected to a paired T-test with a confidence level of 99.5% and 29 degrees of freedom. According to a one-tail T-test table, we need to compare our calculated t-value with the value of 1.699.

Test Number	Number of Tokens	Amir's Time	Colleague's Time	Difference
1	4	38.39362	48.36984	9.97622
2	4	39.11367	49.65975	10.54608
3	4	38.53369	49.32698	10.79329
4	4	37.20625	50.14569	12.93944
5	4	38.91458	49.78653	10.87195
6	4	38.01459	50.96487	12.95028
7	4	39.25478	52.46975	13.21497
8	4	38.65879	51.45698	12.79819
9	4	40.74986	51.24689	10.49703
10	4	39.78963	50.79354	11.00391
11	4	40.35689	53.23698	12.88009
12	4	38.96847	52.34975	13.38128
13	4	38.01465	52.65439	14.63974
14	4	39.11569	53.01154	13.89585
15	4	39.78964	53.24615	13.45651
16	6	54.95987	84.65280	29.69292513
17	6	55.27446	85.44914	30.17467825
18	6	52.36114	84.46955	32.10841023
19	6	54.49469	85.05623	30.56153648
20	6	53.37611	85.26286	31.8867535
21	6	54.61364	85.14409	30.53044707
22	6	58.05734	84.78617	26.72882962
23	6	59.33492	84.36953	25.03460506
24	6	55.13069	85.04686	29.91617443
25	6	59.19231	84.77364	25.58132561
26	6	55.94048	84.87707	28.93659311
27	6	55.06786	84.82687	29.75901321
28	6	54.45313	84.47823	30.02509809
29	6	55.04985	84.07435	29.02449668
30	6	54.39741	84.99850	30.60108675
31	8	78.26205	116.657028	38.39497427
32	8	75.80831	116.31924	40.51092949
33	8	75.55868	116.99526	41.43657589
34	8	78.65301	116.06432	37.4113082
35	8	72.04576	118.91829	46.87252985
36	8	77.80490	115.75485	37.94995174
37	8	76.69070	116.05548	39.36478018
38	8	74.82334	114.75795	39.93460768
39	8	74.27251	113.37381	39.10130376
40	8	70.11775	116.49312	46.37536912
41	8	74.89910	116.00306	41.10396041
42	8	76.60168	119.4113	42.80962497
43	8	79.25859	116.68213	37.42354002
44	8	79.32709	120.26223	40.93513711
45	8	78.25032	116.50902	38.25870219
Mean		56.8656113	84.24983573	27.3842244
Standard Deviation		15.4997172	27.05180664	11.95966262
SE		2.31056141	4.032645237	1.78284124
t		24.6111663	20.89195324	15.35987826
P-value		4.6715E-27	3.61337E-24	5.435E-19

Figure 1: Result of tests

On the condition that the t value that we obtained is more than 1.699, the null hypothesis (H_0) is rejected and the alternative hypothesis (H_a) is accepted. As can be observed, the t value that we obtained is far higher than 1.699, which demonstrates that the null hypothesis is not supported and demonstrates that my code is significantly more efficient than other code in terms of finishing this work.